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Applied Analytics Project - Week 8 Assignment

Model Approach:

For Week 8, our group expanded our modeling approach by testing multiple advanced machine learning models rather than focusing on a single model type.

The models evaluated were:

- HistGradientBoosting
- GradientBoosting
- Random Forest
- Extra Trees
- AdaBoost
- MLP (Neural Network)
- SVC

This decision aligns with our prediction objective of predicting NCAA tournament outcomes because:

- The dataset has many features with complex relationships
- Different models capture different types of patterns
- Comparing multiple model families allows us to identify the best-performing approach

Model Complexity:

The models used this week are significantly more complex than what we have used so far. Boosting models such as HistGradientBoosting and GradientBoosting build trees sequentially where each new tree attempts to correct the errors of the previous ones. Ensemble models like Random Forest and Extra Trees combine predictions from many decision trees to improve overall performance. Neural networks (MLP) introduce layered transformations of the data which allows the model to learn more complex patterns, while SVC models create non-linear decision boundaries to separate classes better.

Compared to Week 6, which used a simple and Logistic Regression model, and Week 7, which introduced a moderately complex Random Forest model, Week 8 implements multiple highly complex model types. This increase in complexity improves the model's ability to capture non-linear patterns in the data, but it also increases the risk of overfitting.

Hyperparameters Evaluated:

This week instead of tuning one model, we performed a broad model comparison across **304 sampled configurations**.

We tested 7 different model families with the following sampled configurations:

- HistGradientBoosting: 60 configs
- Extra Trees: 50 configs
- Random Forest: 50 configs
- Gradient Boosting: 40 configs
- AdaBoost: 32 configs
- MLP: 40 configs
- SVC: 32 configs

We also tested classification thresholds: 0.45, 0.50, 0.55. This is important because it directly affects the tradeoff between precision and recall while allowing us to optimize predictions beyond the default cutoff at 0.5.

Model Performance Metrics:

We used the same evaluation metrics as previous weeks:

- Accuracy: overall prediction correctness
- Precision: limits false win predictions
- Recall: makes sure we capture true wins

These metrics are still appropriate because the problem is binary classification where we care about both prediction accuracy and decision quality.

Model Variations:

For this week, we evaluated 304 model configurations across 7 model families, each tested with 3 thresholds. Each configuration was trained on seasons 2003–2021 and validated using the 2022 season.

We selected the top three best-performing models by ranking all configurations based on validation accuracy and choosing the three with the highest validation performance.

Discuss and analyze how training and validation metrics change across variations:

Across the top three configurations we see very high performance on the training data but lower performance on the validation data. This indicates overfitting. For example, the HistGradientBoosting models achieve perfect training accuracy, precision, and recall (all equal to 1.00), while validation accuracy drops to around 0.81 with lower precision and recall. This gap shows that the models are fitting the training data too closely.

This happens because the models used this week are very flexible and can learn complex patterns. As they continue to refine their predictions, they begin to also capture noise in the training data. This leads to near-perfect training performance but weaker generalization to new data.

We see overfitting in the training data but not in the validation data because the model is trained directly on the training set and can essentially memorize it. The validation set is unseen during training, so it reveals how well the model actually generalizes.

	configId	modelFamily	threshold	fitSeconds	model	set	accuracy	precision	recall
72	12	histGradientBoosting	0.45	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
74	12	histGradientBoosting	0.50	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
76	12	histGradientBoosting	0.55	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
73	12	histGradientBoosting	0.45	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.809524	0.804348	0.925000
75	12	histGradientBoosting	0.50	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.809524	0.804348	0.925000
77	12	histGradientBoosting	0.55	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.793651	0.800000	0.900000
240	40	histGradientBoosting	0.45	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
242	40	histGradientBoosting	0.50	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
244	40	histGradientBoosting	0.55	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	train	1.000000	1.000000	1.000000
241	40	histGradientBoosting	0.45	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.809524	0.791667	0.950000
243	40	histGradientBoosting	0.50	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.793651	0.787234	0.925000
245	40	histGradientBoosting	0.55	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.793651	0.800000	0.900000
1080	180	gradientBoosting	0.45	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	train	0.872633	0.845652	0.992347
1082	180	gradientBoosting	0.50	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	train	0.890706	0.880649	0.969388
1084	180	gradientBoosting	0.55	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	train	0.896730	0.917085	0.931122
1081	180	gradientBoosting	0.45	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	val	0.730159	0.716981	0.950000
1083	180	gradientBoosting	0.50	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	val	0.809524	0.791667	0.950000
1085	180	gradientBoosting	0.55	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	val	0.746032	0.772727	0.850000

Compare performance metrics for the validation dataset across all 3 variations:

	configId	modelFamily	threshold	fitSeconds	model	set	accuracy	precision	recall
241	40	histGradientBoosting	0.45	1.332144	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.809524	0.791667	0.950
1083	180	gradientBoosting	0.50	2.089489	{'modelFamily': 'gradientBoosting', 'randomSta...	val	0.809524	0.791667	0.950
73	12	histGradientBoosting	0.45	1.315345	{'modelFamily': 'histGradientBoosting', 'rando...	val	0.809524	0.804348	0.925

- All models have identical validation accuracy (0.8095)
- Config 12 has the highest precision (0.8043)
- Configs 40 and 180 have the highest recall (0.950)

Overall, the differences between models are very small.

Identify the best model for the week and discuss why you selected that model:

The best model selected for Week 8 was configId 40 (HistGradientBoosting with threshold = 0.45). Both config 40 and config 180 achieved identical validation performance, with accuracy (0.8095), precision (0.7917), and recall (0.95). Since the two models were tied across all key metrics, our code selected config 40 as the top model because it appeared first after sorting by accuracy, recall, and precision.

```
Best model (by validation accuracy, then recall, precision):
configId                                                     40
modelFamily                                histGradientBoosting
threshold                                                  0.45
fitSeconds                                             1.332144
model          {'modelFamily': 'histGradientBoosting', 'rando...
set                                                         val
accuracy                                               0.809524
precision                                              0.791667
recall                                                     0.95
Name: 241, dtype: object
{'modelFamily': 'histGradientBoosting', 'randomState': 226, 'minSamplesLeaf': 20, 'maxLeafNodes': 15, 'maxIter': 600, 'maxDepth': None, 'learningRat
e': 0.08, 'l2Regularization': 0.1, 'threshold': 0.45}
```